

A Synthetic Research Paper for Testing Math Bounding Boxes

MathBox Test Suite

Abstract

This synthetic paper contains representative mathematical constructions at different locations in a two-column document. It is intended for testing the detection and placement of equation bounding boxes rather than presenting a scientific result.

1 Inline and display mathematics

An inline equation such as $E = mc^2$ should be detected without including the surrounding sentence. Multiple expressions can also occur on one line: $a = b$, $x_1 + x_2 = 1$, and $f(t) = e^{-t}$ are three separate inline expressions.

An unnumbered display equation tests centered mathematics:

$$f(x) = x^2 + 2x + 1.$$

The following numbered equation contains subscripts and superscripts:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i. \quad (1)$$

Equation (1) is followed by ordinary prose so that the vertical extent of the equation can be distinguished from adjacent text.

2 Fractions and integrals

Fractions of different heights are useful for checking both the height above the baseline and the depth below it:

$$p(x) = \frac{x^2 + 1}{x^3 - x + 1}. \quad (2)$$

Here is an integral with limits and a nested fraction:

$$I_n = \int_0^1 \frac{x^n}{1 + x^2} dx. \quad (3)$$

Inline fractions provide a different baseline case: the value $r_i = \frac{u_i - v_i}{1 + v_i^2}$ appears in the middle of this sentence.

3 Aligned equations

The `align` environment produces several numbered rows whose alignment points do not necessarily share the

same horizontal extent:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}, \quad (4)$$

$$\nabla \cdot \mathbf{B} = 0, \quad (5)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}. \quad (6)$$

The next display is deliberately placed near the bottom of the first page. The flexible space pushes it down while keeping the source easy to modify.

$$q_{\text{bottom},1} = \sum_{k=1}^n \frac{k}{n^2}. \quad (7)$$

4 Matrices and systems

A matrix tests tall delimiters and entries extending above and below a normal text baseline:

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}. \quad (8)$$

A piecewise definition is another useful tall construction:

$$g(x) = \begin{cases} x^2, & x \geq 0, \\ -x, & x < 0. \end{cases} \quad (9)$$

The following aligned system is unnumbered:

$$\begin{aligned} 2x + 3y - z &= 7, \\ x - y + 4z &= 2, \\ 3x + 2y + 2z &= 9. \end{aligned}$$

5 Long and nested expressions

This equation occupies much of a column's width:

$$\mathcal{L}(\theta) = - \sum_{i=1}^N [y_i \log p_i + (1 - y_i) \log(1 - p_i)]. \quad (10)$$

Nested constructions test whether a box includes every level:

$$h(x) = \sqrt{1 + \frac{x_1^2}{1 + \frac{x_2^2}{1 + x_3^2}}}. \quad (11)$$

Two displays can also appear close together:

$$\begin{aligned} u(t) &= u_0 + at, \\ s(t) &= s_0 + u_0 t + \frac{1}{2} at^2. \end{aligned}$$

The intervening prose helps reveal whether nearby boxes overlap or are merged. For an inline stress test, consider α_i , β_{ij} , $\gamma^{(k)}$, and $T_{\mu\nu}$ within the same line of text.

$$q_{\text{bottom},2} = \lim_{m \rightarrow \infty} \left(1 + \frac{1}{m}\right)^m = e. \quad (12)$$

6 Page-boundary stress tests

The final page combines several forms that are common in technical papers. First, a Fourier transform contains an infinite integral and complex exponential:

$$\widehat{f}(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt. \quad (13)$$

Next, an optimization problem uses constraints and multiple aligned rows:

$$\underset{\mathbf{x}}{\text{minimize}} \quad \frac{1}{2} \mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{c}^\top \mathbf{x} \quad (14)$$

$$\text{subject to} \quad \mathbf{A} \mathbf{x} = \mathbf{b}, \quad (15)$$

$$x_i \geq 0 \quad (i = 1, \dots, n). \quad (16)$$

A short expression $P(A \mid B) = P(A \cap B)/P(B)$ sits between displays. The next equation tests large operators and deeply placed subscripts:

$$Z_N = \prod_{j=1}^N \left(\sum_{k_j=0}^{\infty} e^{-\lambda_j k_j} \right). \quad (17)$$

The last equation is pushed toward the bottom of the final page. This is useful for detecting clipping and off-by-one-page errors in coordinate conversion.

$$\boxed{R_{\text{final}} = \int_0^{2\pi} \int_0^1 r^2 \sin \theta \, dr \, d\theta} \quad (18)$$